



Lakshya JEE 2.0 (2024)

Definite Integration

DPP-06

- The value of $\lim_{x \rightarrow 0} \frac{2 \int_1^{\cos x} \cos^{-1}(t) dt}{2x - \sin 2x}$
 - 0
 - $\frac{1}{2}$
 - $-\frac{1}{2}$
 - $\frac{2}{3}$
- The points of extremum of $\phi(x) = \int_1^x e^{-t^2/2} (1-t^2) dt$ are:
 - $x = 1, -1$
 - $x = -1, 2$
 - $x = 2, 1$
 - $x = -2, 1$
- If $f(x) = \int_2^{x^2} \frac{dt}{1+\sqrt{t}}$, $x > 0$ find $f'(9)$
- Find $\lim_{x \rightarrow 0} \frac{f'(x)}{x}$ where $f(x) = \int_0^x \frac{\sin^2\left(\frac{t}{2}\right)}{t} dt$
- Find $\int_{1/e}^e \frac{1}{x} \sin\left(x - \frac{1}{x}\right) dx$
- If $I_n = \int_0^\pi x^n \sin x dx$ find $I_5 + 20 I_3$.
- Let $f(x) = \int_1^x \frac{\ln t}{1+t} dt$, for $x > 0$, then the value of $f(e) + f\left(\frac{1}{e}\right)$ is
 - 1
 - $\frac{1}{2}$
 - $\frac{1}{4}$
 - None of these
- If $f(x)$ is a continuous function and $\int_{x^2}^{x^4} t^3 f(t) dt = 3 \sin 2\pi x$, then $f(1)$ is equal to
 - 1
 - 1
 - 3π
 - $-\pi$
- If $\int_{\frac{\pi}{2}}^x \sqrt{3-x \sin^2 t} dt + \int_0^y \cos t dt = 0$ then, $\left(\frac{dy}{dx}\right)_{(\pi, \pi)}$ is equal to
 - 3
 - 0
 - $\sqrt{3}$
 - None of these
- If $\int_{\sin x}^1 t^2 f(t) dt = 1 - \sin x$, $x \in \left(0, \frac{\pi}{2}\right)$ then $f\left(\frac{1}{\sqrt{3}}\right)$ equal to
 - 3
 - $\frac{1}{3}$
 - $\frac{1}{\sqrt{3}}$
 - $\sqrt{3}$
- The solution of the integral $\int_0^{2\pi} \sin^4 x \cdot \cos^6 x dx$, is
 - $\frac{3\pi}{128}$
 - $\frac{\pi}{128}$
 - $\frac{5\pi}{128}$
 - $\frac{7\pi}{128}$
- If $F(x) = \frac{1}{x^2} \int_4^x \{4t^2 - 2F'(t)\} dt$, then $F'(4)$ equals
 - 32/9
 - 64
 - 64/9
 - None of these

13. If $F(x) = \int_1^x f(t)dt$, where $f(t) = \int_1^{t^2} \frac{\sqrt{1+u^4}}{u} du$,

then the value of $F''(2)$ equals

- (1) $\frac{7}{4\sqrt{17}}$ (2) $\frac{15}{\sqrt{17}}$
 (3) $\sqrt{257}$ (4) $\frac{15\sqrt{17}}{68}$

14. Let $f: (0, \infty) \rightarrow \mathbb{R}$ be a continuous function such that

$f(x) = \int_0^x t f(t) dt$. If $F(x^2) = x^4 + x^5$, then

$\sum_{r=1}^{12} f(r^2)$ is equal to

- (1) 216 (2) 219
 (3) 222 (4) 225



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

- | | | | |
|----|---------------|-----|-----|
| 1. | (3) | 8. | (3) |
| 2. | (1) | 9. | (3) |
| 3. | $\frac{9}{5}$ | 10. | (1) |
| 4. | $\frac{1}{4}$ | 11. | (1) |
| 5. | (0) | 12. | (1) |
| 6. | π^5 | 13. | (3) |
| 7. | (2) | 14. | (2) |



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>